AI-Powered Personalised Learning Path Generator

Project Title: AI-Powered Personalised Learning Path Generator
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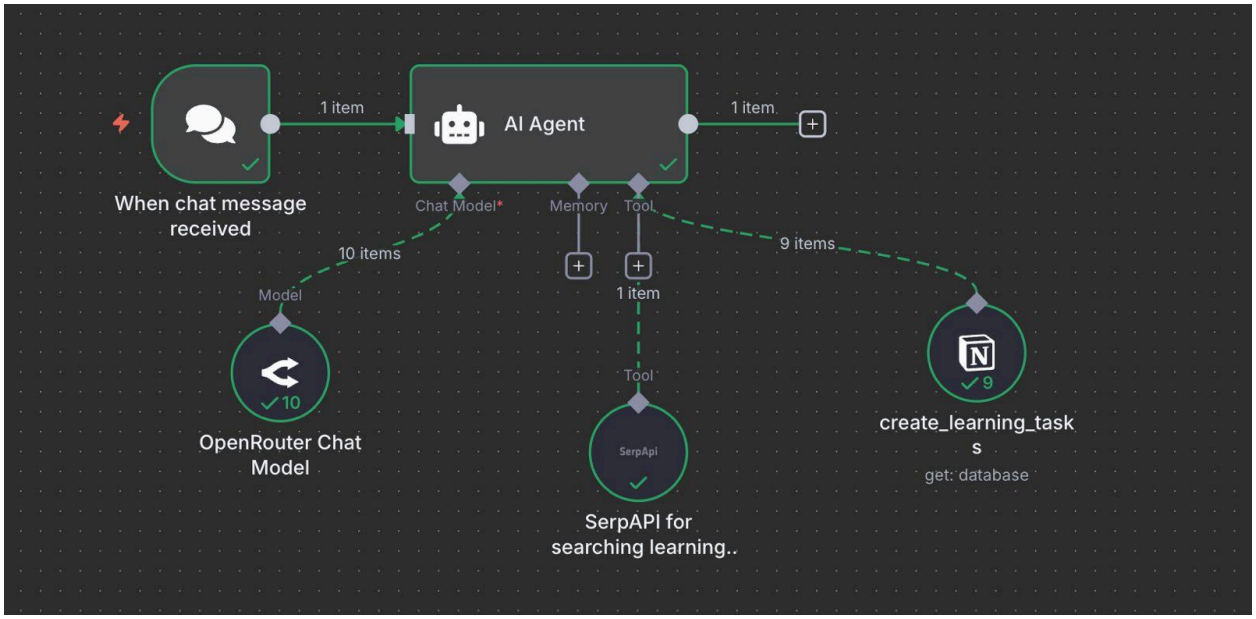
N8N Workflow Screenshot:



N8N Video Demonstration:

https://drive.google.com/file/d/1XitvWee_9eFSG1dEFYShffnW5nfGzxPv/view?usp=drive_link

N8N Workflow JSON File:


```
{
 "name": "Personalised Learning Path Creator",
 "nodes": [
   {
     "parameters": {
       "public": true,
       "initialMessages": "Hi there! 👋\nMy name is Priyanka. What do you want to
learn today?",
       "options": {}
     },
     "type": "@n8n/n8n-nodes-langchain.chatTrigger",
     "typeVersion": 1.3,
     "position": [
       -448,
       0
     ],
     "id": "87ca223d-8c95-4099-a44c-bbea9e406522",
     "name": "When chat message received",
     "webhookId": "0344d81d-644a-4b99-9422-374ac8cf4c21"
   },
   {
     "parameters": {
       "promptType": "define",
       "text": "=Create a personalized learning path for: {{ $json.chatInput
}}\n\nIMPORTANT:\n- First use the search_learning_resources tool\n- Then store each
learning step in Notion using the create_learning_tasks tool",
       "options": {
         "systemMessage": "=You are an AI learning assistant.\n\nYour goal is to
create a structured learning path for a given skill and store it in a Notion task
tracker database.\n\nSTRICT RULES:\n1. You MUST use the tool
\"search_learning_resources\" exactly ONCE.\n2. Do NOT call the search tool more than
once.\n3. After retrieving results, analyze them and create a learning plan.\n4. Do
NOT generate final output as text.\n5. You MUST use the tool \"create_learning_tasks\"
to store each learning step.\n\nTOOL USAGE:\n- Each learning step must be added as a
separate task.\n- Call \"create_learning_tasks\" once per step.\n\nTOOL INPUT FORMAT
(VERY IMPORTANT):\n\nWhen calling the tool, you MUST fill in all the Notion (tool
\"create_learning_tasks\") database (url:
\"https://www.notion.so/32f50c3944b980318974f795c69159ff?v=32f50c3944b98049a1fc000c80f
f4f5a&source=copy_link\") columns:\n{\n  \"Task name\": \"Sub-topic name\",\n
\"Status\": \"Not Started\",\n  \"Assignee\": \"Me\",\n  \"Due date\":
\"DD-MM-YYYY\",\n  \"Priority\": \"Low | Medium | High\",\n  \"Description\": \"Clear
```

```
explanation of the learning task\",\n  \"Effort level\": \"Small | Medium |
Large\"\n}\n\nFIELD GUIDELINES:\n- Task name → actual topic (NOT Step 1)\n-
Description → clear learning instruction\n- Due date → in relative format like it
could be within the next few weeks\n- status → ALWAYS \"Not Started\"\n- Priority:\n
- High → important foundational steps\n  - Medium → standard learning steps\n  - Low →
optional or advanced topics\n- Effort level:\n  - Large → complex topics (e.g., deep
learning, projects)\n  - Medium → moderate difficulty\n  - Small → basic concepts\n-
assignee → ALWAYS \"Me\"\n\nOTHER RULES:\n- Create 8-10 steps maximum\n- Do NOT loop
or repeat tool calls\n- Do NOT skip tool usage\n\nIMPORTANT:\n- Do NOT return JSON as
final output\n- Do NOT return plain text\n- ONLY use tools to complete the task"
      }
    },
    "type": "@n8n/n8n-nodes-langchain.agent",
    "typeVersion": 2.2,
    "position": [
      -240,
      0
    ],
    "id": "da6081ff-7d18-4845-90b5-a60576c6e464",
    "name": "AI Agent"
  },
  {
    "parameters": {
      "model": "nvidia/nemotron-3-nano-30b-a3b:free",
      "options": {}
    },
    "type": "@n8n/n8n-nodes-langchain.lmChatOpenRouter",
    "typeVersion": 1,
    "position": [
      -384,
      208
    ],
    "id": "4b5ef244-a54f-4096-830b-3ef31be2e704",
    "name": "OpenRouter Chat Model",
    "credentials": {
      "openRouterApi": {
        "id": "YyXskfaF0bjcH0tZ",
        "name": "OpenRouter account"
      }
    }
  },
  {
```

```
    "parameters": {
      "options": {}
    },
    "type": "@n8n/n8n-nodes-langchain.toolSerpApi",
    "typeVersion": 1,
    "position": [
      -96,
      272
    ],
    "id": "a77a60ff-acab-427b-abe9-d64f28cd58a2",
    "name": "SerpAPI for searching learning resources",
    "credentials": {
      "serpApi": {
        "id": "UygLLzXPDqYOcXaP",
        "name": "SerpAPI account 3"
      }
    },
    "notes": "Use this tool to search for tutorials, courses, and learning
resources."
  },
  {
    "parameters": {
      "descriptionType": "manual",
      "toolDescription": "Create existing tasks in a Notion task tracker
database.\n\nThis tool is used to modify and populate all relevant fields of a task,
including task name, description, status, priority, assignee, due date, and effort
level.\n\nEach update represents one step in a personalized learning path, ensuring
that all task properties are accurately filled and aligned with the learning
workflow.",
      "resource": "databasePage",
      "databaseId": {
        "__rl": true,
        "value": "32f50c39-44b9-8031-8974-f795c69159ff",
        "mode": "list",
        "cachedResultName": "Tasks Tracker",
        "cachedResultUrl": "https://www.notion.so/32f50c3944b980318974f795c69159ff"
      },
      "title": "={{ $json.chatInput }}",
      "propertiesUi": {
        "propertyValues": [
          {
            "key": "Task name|title",
```

```
            "title": "={{ /*n8n-auto-generated-fromAI-override*/
$fromAI('propertyValues0_Title', ``, 'string') }}"
          },
          {
            "key": "Description|rich_text",
            "textContent": "={{ /*n8n-auto-generated-fromAI-override*/
$fromAI('propertyValues1_Text', ``, 'string') }}"
          },
          {
            "key": "Due date|date",
            "includeTime": false,
            "date": "2026-04-29T14:31:35",
            "timezone": "Asia/Kolkata"
          },
          {
            "key": "Effort level|select",
            "selectValue": "Medium"
          },
          {
            "key": "Status|status",
            "statusValue": "Not started"
          },
          {
            "key": "Assignee|people",
            "peopleValue": [
              "9fe974ed-ad09-45c6-8fce-ffcdd934d27c"
            ]
          },
          {
            "key": "Priority|select",
            "selectValue": "High"
          }
        ]
      },
      "options": {}
    },
    "type": "n8n-nodes-base.notionTool",
    "typeVersion": 2.2,
    "position": [
      176,
      192
    ],
```

```
    "id": "00822b2f-b5bd-4a0d-9ec4-88b44781b3a3",
    "name": "create_learning_tasks",
    "credentials": {
      "notionApi": {
        "id": "PsgtIwNsmXCXa81f",
        "name": "Notion account 6"
      }
    },
    "notes": "Create tasks in a Notion database with step details, priority, effort
level, and assignment information."
  }
],
"pinData": {},
"connections": {
  "When chat message received": {
    "main": [
      [
        {
          "node": "AI Agent",
          "type": "main",
          "index": 0
        }
      ]
    ]
  },
  "OpenRouter Chat Model": {
    "ai_languageModel": [
      [
        {
          "node": "AI Agent",
          "type": "ai_languageModel",
          "index": 0
        }
      ]
    ]
  },
  "SerpAPI for searching learning resources": {
    "ai_tool": [
      [
        {
          "node": "AI Agent",
          "type": "ai_tool",
```

```
          "index": 0
        }
      ]
    ]
  },
  "create_learning_tasks": {
    "ai_tool": [
      [
        {
          "node": "AI Agent",
          "type": "ai_tool",
          "index": 0
        }
      ]
    ]
  }
},
"active": false,
"settings": {
  "executionOrder": "v1"
},
"versionId": "723d81f7-a604-421f-90be-3776b03801d2",
"meta": {
  "templateCredsSetupCompleted": true,
  "instanceId": "c76cbd41aac0dfc1d3fc04431354ea8db1c7d28d2aab6034aa40ae73cdec24d6"
},
"id": "sGwBmoO0cbAhtG9G",
"tags": []
}
```

Introduction

In today's fast-paced digital world, learning new skills has become essential for personal and professional growth. However, identifying relevant resources, structuring a learning roadmap, and maintaining consistency can be time-consuming and often overwhelming. Typically, learners spend a significant amount of time searching through multiple websites, comparing tutorials, and organising content into a meaningful sequence.

To address this challenge, I developed an AI-powered workflow that automates the creation of personalised learning paths. The system takes a user-defined skill as input and generates a structured, step-by-step learning plan. By integrating artificial intelligence with workflow automation, this project serves as a virtual learning assistant that simplifies the planning and organisation of educational content.

The motivation behind this project was to reduce manual effort, improve efficiency, and demonstrate the practical application of AI agents in real-world problem-solving.

Workflow Design

The workflow for this project was built with n8n, which enables seamless integration with AI models and external tools.

The process begins with a chat trigger node where the user inputs the desired skill, such as "Machine Learning." This input is then passed to the AI Agent, which serves as the core component of the workflow. The AI Agent is powered by a chat model integrated through OpenRouter, enabling it to understand user intent and generate structured outputs.

To gather relevant information, the agent uses a web search tool powered by SerpAPI. This tool retrieves high-quality learning resources from the internet, ensuring that the generated learning path is accurate and up-to-date.

After collecting the necessary data, the AI Agent processes the information and creates a structured set of learning tasks. Each task represents a sub-topic rather than a generic step, making the learning path more practical and actionable. The agent also assigns attributes such as priority and effort level based on the complexity and importance of each topic.

The final step is to store these tasks in a Notion database using the Notion integration. Each entry in the database includes fields such as task name, description, status, priority, effort level, assignee, due date, and a YouTube video link for additional learning support. This transforms the output into a usable task tracker, enabling users to follow and manage their learning journey effectively.

Challenges & Solutions

During the development of this project, several technical challenges were encountered.

One of the primary challenges was ensuring that the AI Agent correctly utilised external tools. Initially, the agent generated responses without calling the search tool, leading to less accurate outputs. This issue was resolved by refining the system prompt and tool descriptions to explicitly instruct the agent on when and how to use each tool.

Another significant challenge was handling schema mismatches between the AI output and the Notion database. When too many fields were passed directly from the AI, the workflow produced errors due to mismatches in format. To solve this, the output structure was simplified, and additional fields were handled within the workflow configuration, ensuring compatibility and stability.

API authentication also posed difficulties, particularly while integrating external services. Proper configuration of API keys and permissions was necessary to establish successful connections between the workflow and external tools.

Additionally, managing structured outputs such as priority levels and effort estimation required careful prompt engineering. By clearly defining rules and constraints within the system message, the agent generated consistent, meaningful outputs.

Conclusion

This project successfully demonstrates how AI agents can be integrated with workflow automation to solve real-world problems. By automating the process of creating personalised learning paths, the system significantly reduces manual effort and enhances productivity.

The integration of AI reasoning, web search, and task management tools highlights the potential of intelligent automation in education and skill development. The project not only showcases technical implementation but also emphasises the importance of prompt engineering and system design in building reliable AI-driven workflows.

In the future, this system can be further enhanced by integrating additional APIs, improving recommendation quality, and developing a user-friendly interface. Overall, the project reflects the growing impact of AI in transforming how we learn and interact with information.